# Report 2: Use of AI Tools

Information and Coding 2026/27

Dinis Oliveira André Silva Rafael Ferreira

25 October 2026

## 1 AI Tools Used

We used one tool, Claude (Anthropic), in chat and in Claude Code, mainly like a search engine for solutions: for each problem we asked for the known approaches, their pros and cons, and a recommendation. **The decision on every piece of code and every codec choice was ours.** Where Claude drafted code, it did so in small steps for a design we had chosen, and we read, ran and changed it. It could not run our benchmarks, so all numbers in Report 1 are from our own runs.

## 2 Cases

**Lossless codec.** We asked for the options at each stage. Stereo: independent channels or mid/side. Prediction: fixed polynomial predictors (fast, no coefficients) or adaptive linear prediction as in `flac` (compresses more, slower). Entropy coding: Rice (simple, near-optimal) or arithmetic coding (smaller, slower). We chose mid/side, fixed predictors and Rice for speed and simplicity, and a verbatim copy of the WAV header and trailer so the output is byte-identical. *Verified* with 55 synthetic round-trip cases, 60 damaged files with sanitizers and a byte comparison of all 27 files; our files are 10.5% larger than `flac -8`, the price of the choice. *Wrong claims we caught with data:* DSD100 assumed to be full songs (they are 30 s clips), and our codec said to beat every general-purpose tool (on SQAM, `xz -9e` wins). We changed the block size from 4096 to 512 after a sweep.

**Lossy codec.** Claude explained the concepts and the trade-offs: DCT or MDCT, rate control by step or target SNR, Huffman or adaptive range coding. We chose both transforms (selectable, to compare them), a global step set by `-step` or `-snr`, an adaptive binary range coder and block size 2048. *Verified* with a high-quality round trip, the SNR from `wav_cmp` against the target, and a comparison with MP3 and AAC using `ffmpeg`.

**Report 1.** Claude proposed a page budget, a skeleton, drafts and figure scripts; we decided the content. Recomputing its numbers found errors ("34% smaller than `zstd -19`" is 24%), and it caught our $\Delta/2$ maximum error, which our data show is $\Delta - 1$ because `wav_quant` truncates.

## 3 Critical Assessment

It helped most in comparing options quickly, drafting boilerplate and tests, and catching a text/data mismatch. Like a search result, it stated unverified facts confidently (sizes, the SQAM cause) and slipped on percentages. Our rules: ask for options with trade-offs, decide ourselves, build and test in small steps, recompute every number, and treat any "why" as a hypothesis until measured. It saved time but did not remove the need to understand the work.